UTILIZING ACOUSTIC VIBRATIONS TO NON-INVASIVELY ANALYZE STRUCTURAL DEFORMATION IN BUILDINGS

Introduction

Hollow metal pipes are used in the **foundation of structures for support,** such as columns and beams. In pressing situations such as earthquakes, **buildings** are put at **severe risk** after experiencing such shock.

Roughly **75% of deaths** following earthquakes result from structural collapses, deeming it an issue that requires **immediate attention.** With meaningful progress made, it is likely that **future buildings** in **disaster prone** areas can be **attended to** more **urgently,** conserving structures and **saving lives**.

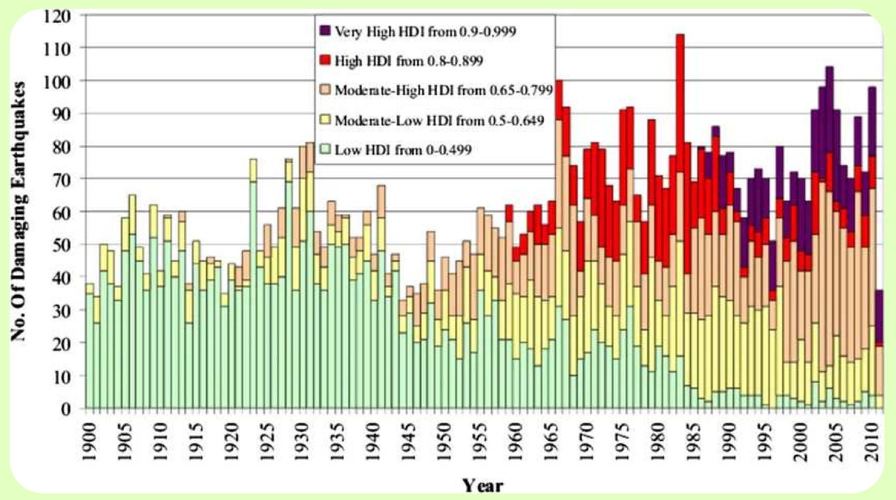

Figure 1: **Earthquake damage overtime**

An **array of metal pipes** will be deformed to various **levels,** each level will have samples **tested with an accelerometer** to generate **vibration readings.** The vibration readings will ideally **differ** based on the **intensity** of the **deformation** in each pipe. These readings will be used to create a **machine learning** model that can **identify** the **relative deformation** level of a pipe **given** the **vibration** readings.

Purpose

When there is **seismic activity,** finding the buildings in **most need of attention** is difficult. By analyzing structural deformation using acoustics, a machine learning model can be utilized to identify how **intensely a column is deformed** inside a structure, so that the structures with **stronger deformities** can be attended to.

Methods

1: Cutting and Shaping

- Marked down the middle of a 5' metal pipe into equal 2.5' segments
- Utilized a metal chop saw with a steel blade to cut the pipe in half
- Sanded the edges and deburred the inside to remove sharp imperfections

2: Deformation

- Placed within a 3-roller pipe bender such that the pipe was placed evenly in the middle
- Lowered the top wheel to a preset height (x% deformed) and applied the desired bend
- Repeated the processs for all deformation levels (50 samples per level)

Figure 2: Metal chop saw with marked pipe placed for cutting halfway through

Figure 3: Various pipe deformations

Figure 4: Bending a pipe using the 3 - roller pipe bender to deform to 30%

3: Data Collection

- Wired and coded an LSM9D1 accelerometer to capture x,y,z movements
- Attached the sensor to the pipe and vibrated it by dropping a mass
- Captured a screenshot of the graph from the Arduino serial plotter

4: ML Model + Testing

- Split the data 70% train, 20% validation, 10% test and used a CNN to process the data
- Tested the model and evaluated the accuracy and loss
- Made changes as needed based on the output graphs

Figure 5: Testing Setup

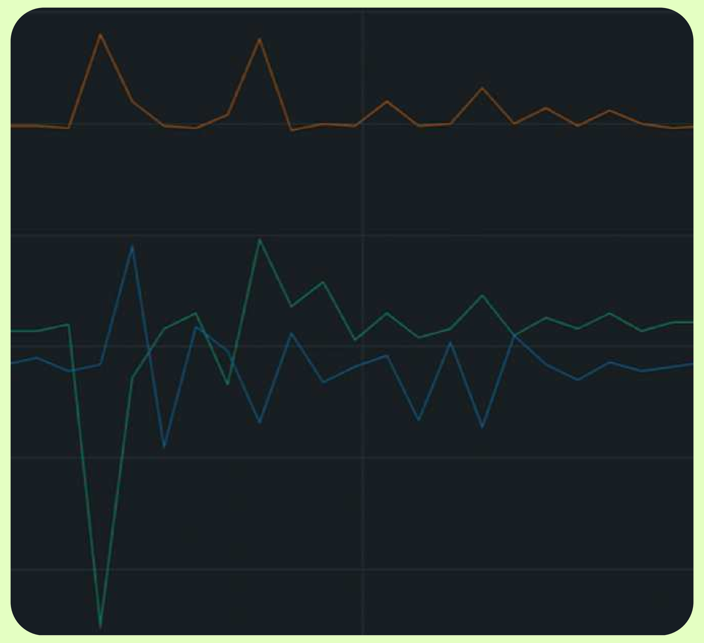

Figure 6: 10% defomed sample

```
import imghdr
import numpy as np
from matplotlib import pyplot as plt
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Conv2D, MaxPooling2D, Dense, Flatten
from tensorflow.keras.metrics import Precision, Recall, BinaryAccuracy
import cv2
from tensorflow.keras.models import load_model
```

Figure 7: ML Packages

```
model = Sequential()
model.add(Conv2D(11, (3,3), 1, activation='relu', input_shape=(256,256,3)))
model.add(MaxPooling2D())
model.add(Conv2D(32, (3,3), 1, activation='relu'))
model.add(MaxPooling2D())
model.add(Conv2D(32, (3,3), 1, activation='relu'))
model.add(MaxPooling2D())
model.add(Flatten())
model.add(Dense(256, activation='relu'))
model.add(Dense(1, activation='sigmoid'))
model.compile('adam', loss=tf.losses.BinaryCrossentropy(), metrics=['accuracy'
```

Figure 8: The CNN

Results

Figure 9: 10% Deformation: High initial z (red), High z amplitude (red), High x frequency

Figure 10: 20% Deformation: Middling initial z (red), Middling z amplitude (red), Middling x frequency

Figure 11: 30% Deformation: Low initial z (red), Low z amplitude (red), Low x frequency

Conclusions + Extensions

There are a couple of **key indicators** that changed with the deformation. One of these indicators was the **z-axis position**. As the deformation increased, it was clear that the z-axis position represented by the **red line in figures 6, 7, and 8 above decreased in value.** The z-values from 10%, 20%, and 30% deformation can be seen across the 3 figures above. Another clear trend was that **the vibrations following** the **immediate release** of the mass steadily **decreased in amplitude on the z-axis** as the deformation intensity increased. Finally, there was **decreased vibration duration** and **intensity** on the **x-axis** represented by the **green line.** Future goals will center around sending **vibrations** through the **pipe** to **model** the **shape** of the deformation.

References

21 injured, dozens of buildings collapse as 5.4 quake hits East China. NDTV. (2023, August 6). https://www.ndtv.com/world-news/21-injured-dozens-of-buildings-collapse-as-5-4-quake-hits-east-china-4274143

Avci, O., Abdeljaber, O., Kiranyaz, S., Hussein, M., Gabbouj, M., & Inman, D. J. (2021). A review of vibration-based damage detection in civil structures: From traditional methods to Machine Learning and Deep Learning applications. Mechanical Systems and Signal Processing, 147, 107077. https://doi.org/10.1016/j.ymssp.2020.107077

Defects of hot-rolled seamless steel pipe in the production process. CONTINENTAL STEEL CO.,LTD. (n.d.). https://www.continental-steel.com/blogs/defects-of-hot-rolled-seamless-steel-pipe-in-the-production-process.html